

Vivado FPGA Flow

Synthesis, Timing Closure, and Implementation on the Nexys A7-100T

Target Device: Xilinx Artix-7 (xc7a100tcsg324-1)

Toolchain: Xilinx Vivado Design Suite

Board: Digilent Nexys A7-100T

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1. FPGA Board Selection

The **Nexys A7-100T** (part **xc7a100tcsg324-1**) was selected for this design based on its available fabric resources and peripheral support. The FPGA provides **63,400** Look-Up Table (LUT) elements, **126,800** Flip-Flops (FFs), **135** Block RAMs (BRAM), **240** DSP slices, and **210** user-available I/O pins. This capacity accommodates the baseband DQPSK transmitter logic, the **2,048-bit** PPDU packet buffering structures, and the high-speed I/Q mapping tables without exceeding hardware limits. The onboard **100 MHz** oscillator and **3.3 V PMOD headers** provide a straightforward interface to external logic analyzers and DAC peripherals.

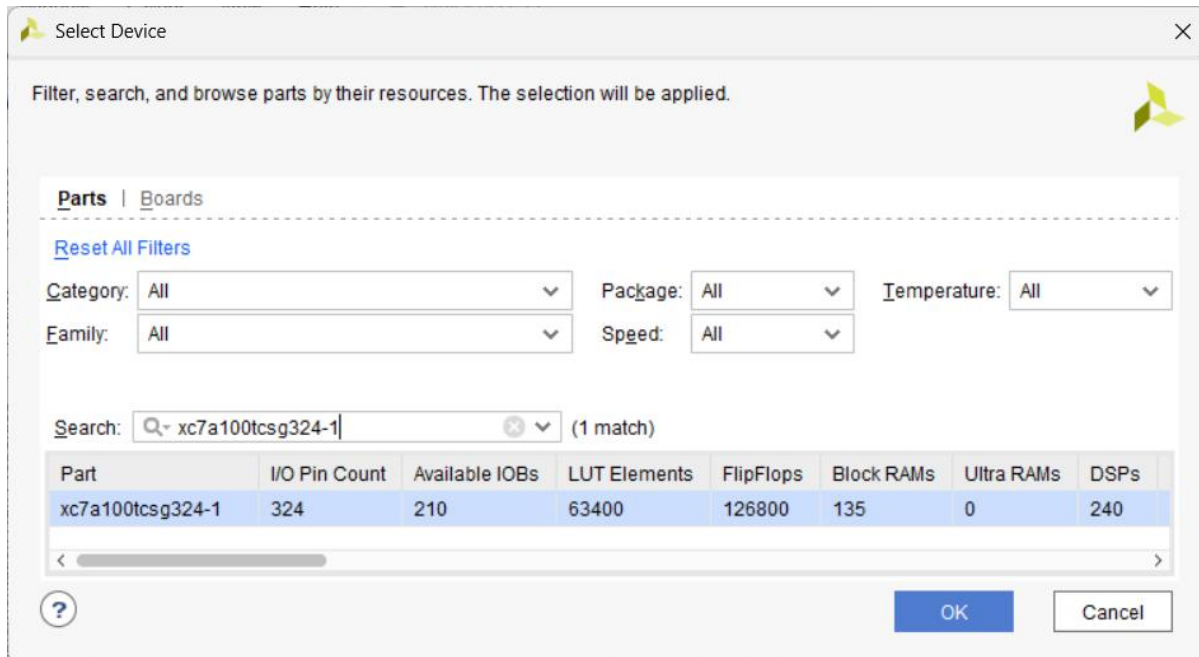


Figure 1. Nexys A7-100T development board

2. Synthesis Issue & RTL Fix

Synthesis hung indefinitely during elaboration due to two independent architectural bottlenecks in **form_ppdu.v** and **css_phy_tx_top.v**.

2.1 Root Cause Analysis

1. **Unrolled Dynamic Write Loop (form_ppdu.v):** The I/Q FIFOs were declared as unpacked arrays (reg `i_fifo [0:2047]`). Inside the write process, a for loop attempted to write up to 64 bits simultaneously using dynamic offset indexing (`i_fifo[wr_ptr + j] <= i_in[j]`). During synthesis elaboration, Vivado fully unrolled this loop across **2,048** individual register instances, building over **130,000** dynamic comparator and decoder terms. This overwhelmed the logic optimization engine and caused synthesis to freeze.

2. **Failed BRAM Inference (css_phy_tx_top.v):** The 1,024-word capture buffers (dqpsk_mem_real and dqpsk_mem_imag) were read asynchronously via combinational logic (assign csk_s_real = dqpsk_mem_real[csk_symbol_index]). Vivado initially attempted to map these memories to Block RAM (BRAM). Because Xilinx BRAM primitives strictly require synchronous clocked reads, BRAM inference failed. Vivado defaulted to dissolving both memory arrays into **4,096** scalar flip-flops driven by massive 1024-to-1 multiplexer trees, creating an explosion in logic complexity.

2.2 RTL Modifications & Fixes

2.2.1 Packed Vector & Indexed Part-Select (form_ppdu.v)

The FIFO structures were converted from unpacked arrays to 2,048-bit packed vector registers (reg [0:2047] i_fifo;). The unrolled for loop was replaced with Verilog indexed part-select statements (i_fifo[wr_ptr +: 4] and i_fifo[wr_ptr +: 64]). This allows Vivado to infer a clean barrel-shifter enable structure in under a second without generating unrolled comparator networks.

Before

```
// Unpacked array declaration
reg i_fifo [0:2047];
reg q_fifo [0:2047];

// Dynamic offset writing inside unrolled loop causes 130k+ comparator terms
integer j;
always @(posedge clk) begin
    for (j = 0; j < bit_count; j = j + 1) begin
        i_fifo[wr_ptr + j] <= i_in[j];
        q_fifo[wr_ptr + j] <= q_in[j];
    end
end
```

After

```
// Packed vector declaration
reg [0:2047] i_fifo;
reg [0:2047] q_fifo;

// Indexed part-select (+:) infers a clean barrel-shifter structure
always @(posedge clk) begin
    i_fifo[wr_ptr +: 64] <= i_in; // Supports indexed slices (+: 4 or +: 64)
    q_fifo[wr_ptr +: 64] <= q_in;
end
```

2.2.2 Distributed LUTRAM Attribute (css_phy_tx_top.v)

The synthesis attribute (* ram_style = "distributed" *) was added directly above dqpsk_mem_real and dqpsk_mem_imag. This forces Vivado to target Distributed LUTRAM, which natively supports asynchronous combinational reads and eliminates the 1024:1 multiplexer expansion completely.

Before

```
// Unattributed memory declaration leading to failed BRAM inference & MUX
expansion
reg [7:0] dqpsk_mem_real [0:1023];
reg [7:0] dqpsk_mem_imag [0:1023];

assign csk_s_real = dqpsk_mem_real[csk_symbol_index]; // Asynchronous read
```

After

```
// Explicit Distributed LUTRAM attribute
(* ram_style = "distributed" *) reg [7:0] dqpsk_mem_real [0:1023];
(* ram_style = "distributed" *) reg [7:0] dqpsk_mem_imag [0:1023];

assign csk_s_real = dqpsk_mem_real[csk_symbol_index]; // Asynchronous read
natively supported
```

3. PLL / 32 MHz Clock

The DQPSK transmitter logic operates at a strict **32 MHz** system clock to support the required target baud rate and to match external DAC sampling frequencies. Because the Nexys A7 oscillator is fixed at **100 MHz**, a Phase-Locked Loop (PLL) via Vivado's Clocking Wizard IP was required. The MMCM (Mixed-Mode Clock Manager) safely steps the 100 MHz reference down to 32 MHz, guaranteeing low jitter, correct phase alignment, and a stable duty cycle.

Re-customize IP

Clocking Wizard (6.0)

Documentation IP Location Switch to Defaults

IP Symbol Resource

Show disabled ports

Component Name: clk_wiz_0

Tab: Clocking Options

The phase is calculated relative to the active input clock

Output Clock	Port Name	Output Freq (MHz)		Phase (degrees)		Duty Cycle (%)	
		Requested	Actual	Requested	Actual	Requested	Actual
☒ clk_out1	clk_out1	32	32.000	0.000	0.000	50.000	50.0
☐ clk_out2	clk_out2	100.000	N/A	0.000	N/A	50.000	N/A
☐ clk_out3	clk_out3	100.000	N/A	0.000	N/A	50.000	N/A
☐ clk_out4	clk_out4	100.000	N/A	0.000	N/A	50.000	N/A
☐ clk_out5	clk_out5	100.000	N/A	0.000	N/A	50.000	N/A
☐ clk_out6	clk_out6	100.000	N/A	0.000	N/A	50.000	N/A

OK Cancel

Figure 2. Vivado Clocking Wizard PLL settings

4. Post-PLL Schematic

In the elaborated design, the physical **E3** pin (CLK100MHZ) routes exclusively into the `clk_in1` port of the Clocking Wizard IP. The resulting `clk_out1` (**32 MHz**) drives the main clock tree, fanning out to the clock pins of every sequential element within the PPDU formatter and DQPSK modulator. The physical reset button drives the PLL's reset input, ensuring no processing occurs during clock acquisition.

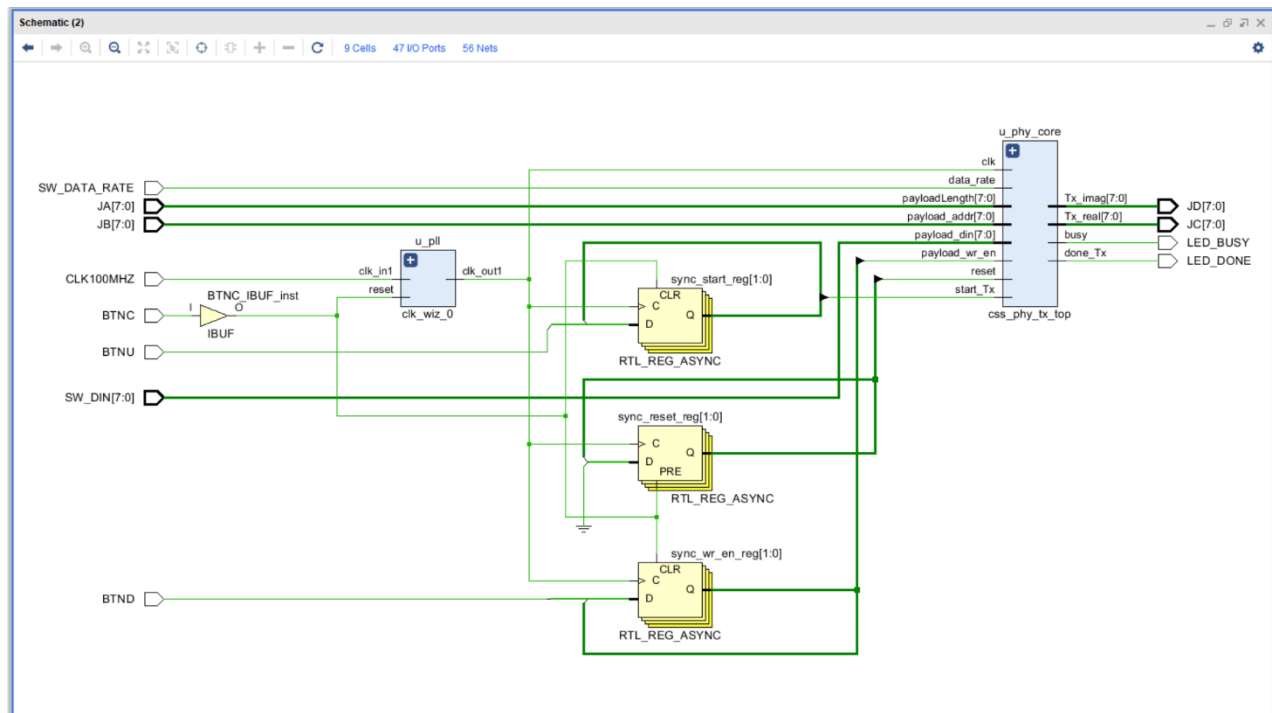


Figure 3. Post-PLL elaborated design schematic

5. Initial XDC Constraints

The initial Master XDC file defined the physical pinouts and I/O standards for the design.

```
set_property -dict { PACKAGE_PIN E3  IOSTANDARD LVCMOS33 } [get_ports { CLK100MHZ }];
create_clock -name sys_clk_pin -period 10.00 -add -waveform {0 5} [get_ports { CLK100MHZ }];

set_property -dict { PACKAGE_PIN J15 IOSTANDARD LVCMOS33 } [get_ports { SW_DIN[0] }];
# ... [Additional Switch/Button Constraints] ...

set_property -dict { PACKAGE_PIN C17 IOSTANDARD LVCMOS33 } [get_ports { JA[0] }];
# ... [Additional PMOD Constraints for JB, JC, JD] ...
```

PMOD ports (**JA**, **JB**, **JC**, **JD**) were specifically chosen for external interfacing because they provide unbuffered 3.3 V LVCMOS connections directly to the FPGA logic fabric. In this design, **JC** and **JD** were mapped to Tx_real and Tx_imag to easily output the 8-bit I/Q baseband signals to standard breadboards or external digital-to-analog converters, while **JA** and **JB** were allocated for dynamic payload configuration.

6. Initial Timing Warnings

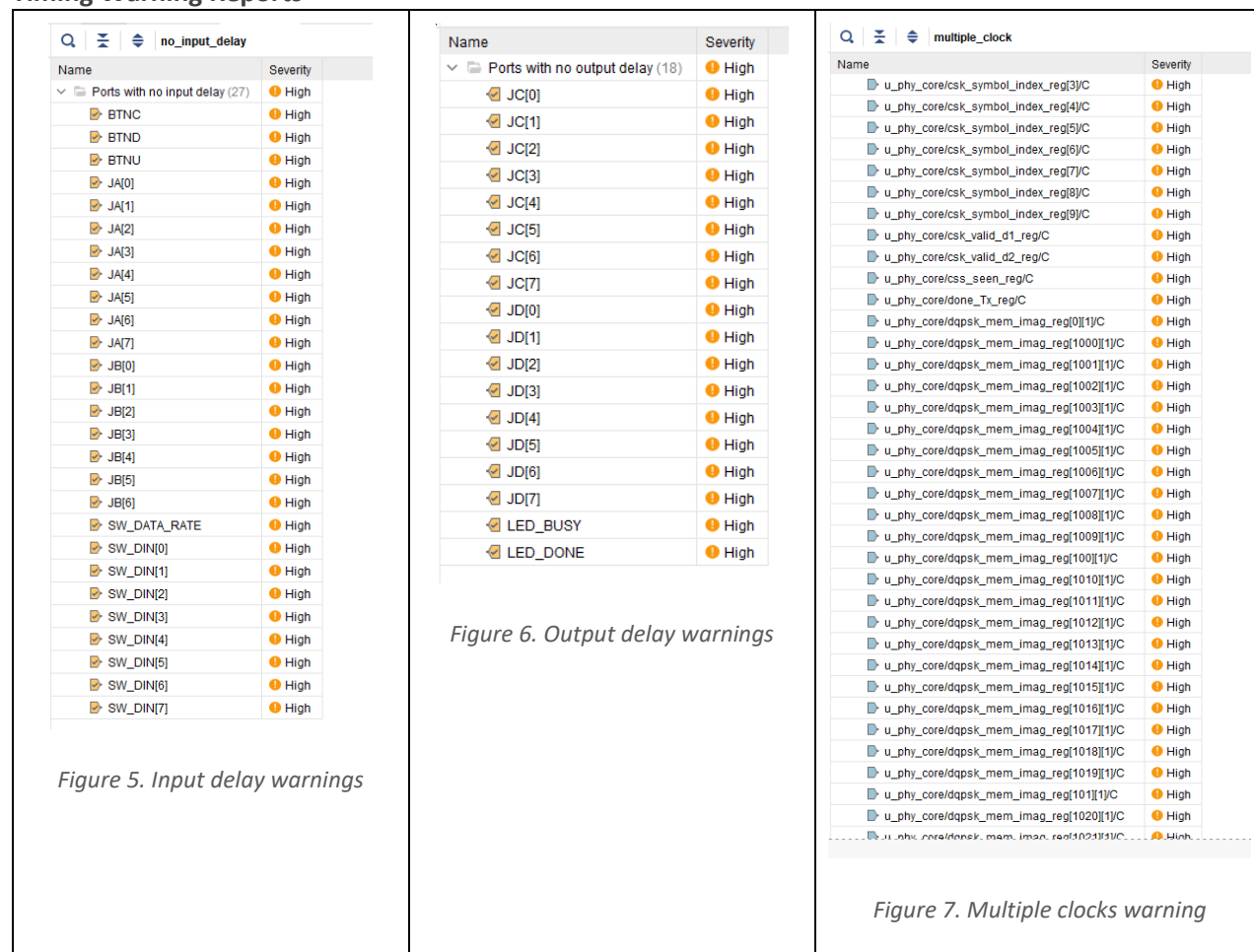
Upon running implementation, Vivado generated a series of critical timing warnings.

- **Multiple Clocks:** Caused by the -add flag in the create_clock command. Vivado's board files automatically apply a 100 MHz definition to pin E3. The -add flag forced Vivado to create a second, overlapping 100 MHz clock on the identical net. Both constraints propagated through the PLL, forcing the timing engine to analyze every internal register twice.
- **No Input / Output Delay:** Static Timing Analysis requires known setup and hold boundaries for external devices. Because the PMODs, switches, and LEDs lacked set_input_delay and set_output_delay constraints, Vivado flagged them as unconstrained endpoints.

Name	Waveform	Period (ns)	Frequency (MHz)
CLK100MHZ	{0.000 5.000}	10.000	100.000
clk_out1_clk_wiz_0	{0.000 15.625}	31.250	32.000
clkfbout_clk_wiz_0	{0.000 25.000}	50.000	20.000
sys_clk_pin	{0.000 5.000}	10.000	100.000
clk_out1_clk_wiz_0_1	{0.000 15.625}	31.250	32.000
clkfbout_clk_wiz_0_1	{0.000 25.000}	50.000	20.000

Figure 4. Clock summary report

Timing Warning Reports



7. XDC Fix

To resolve the timing analysis violations, the XDC file was updated.

```
## Removed the "-add" flag to prevent duplicate clock definitions
set_property -dict { PACKAGE_PIN E3  IOSTANDARD LVCMOS33 } [get_ports { CLK100MHZ }];
create_clock -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports { CLK100MHZ }];

## Instruct timing engine to ignore asynchronous/human inputs and slow PMOD outputs
set_false_path -from [get_ports {BTNC BTNU BTND SW_DATA_RATE SW_DIN[*] JA[*] JB[*]}]
set_false_path -to [get_ports {LED_BUSY LED_DONE JC[*] JD[*]}]
```

Removing the -add flag allowed the primary clock constraint to cleanly overwrite the board default, providing exactly one clock definition for the PLL to inherit. The set_false_path commands were added because the external interfaces (human-operated switches and slow PMOD debug signals) are

asynchronous to the 32 MHz clock. This explicitly commands the Static Timing Analyzer to stop expecting synchronous timing closure on these specific pins.

8. Timing Verification

After regenerating implementation with the revised XDC constraints, the timing analysis reports cleanly passed. The `set_false_path` directives effectively masked the asynchronous I/O, driving the `no_input_delay` and `no_output_delay` violation counts to zero. Furthermore, checking the clock networks confirmed only one master clock drives the PLL, entirely resolving the overlapping constraint collision.

Tcl Console

Messages

Log

Reports

Design Runs

Power

DRC

Methodology

Timing

x

Q

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●

Q

Check Timing

General Information

Timer Settings

Design Timing Summary

Clock Summary (3)

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📁

 Check Timing (45)

no_clock (0)

constant_clock (0)

pulse_width_clock (0)

unconstrained_internal_endpoints (0)

🟡 no_input_delay (27)

🟡 no_output_delay (18)

multiple_clock (0)

generated_clocks (0)

loops (0)

partial_input_delay (0)

partial_output_delay (0)

latch_loops (0)

> 📁 Intra-Clock Paths

Inter-Clock Paths

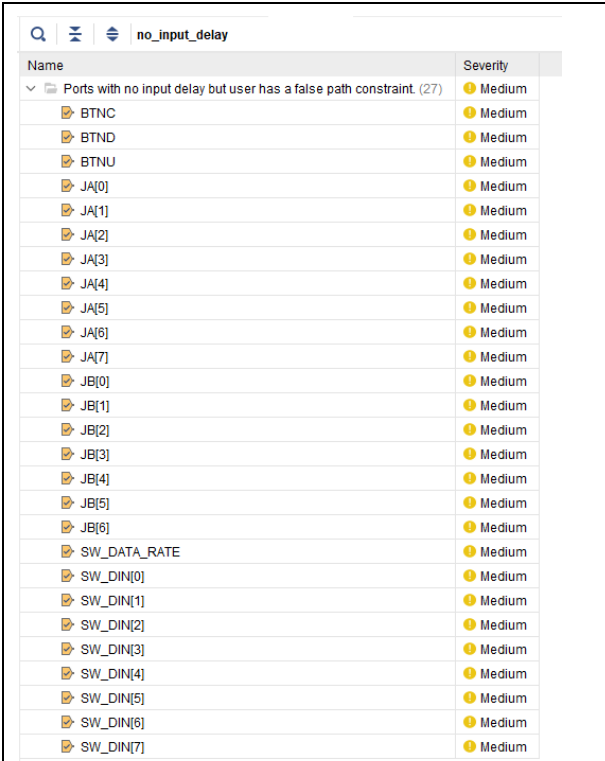
> 📁 Other Path Groups

User Ignored Paths

Unconstrained Paths

Timing Check	Count	Worst Severity
no_input_delay	27	🟡 Medium
no_output_delay	18	🟡 Medium
no_clock	0	
constant_clock	0	
pulse_width_clock	0	
unconstrained_internal_endpoints	0	
multiple_clock	0	
generated_clocks	0	
loops	0	
partial_input_delay	0	
partial_output_delay	0	
latch_loops	0	

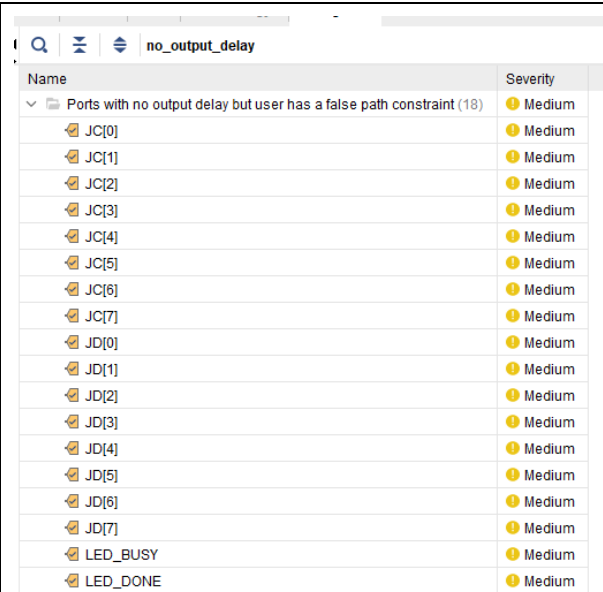
Figure 8. Check timing report — post-fix



no_input_delay

Name	Severity
Ports with no input delay but user has a false path constraint (27)	Medium
BTNC	Medium
BTND	Medium
BTNU	Medium
JA[0]	Medium
JA[1]	Medium
JA[2]	Medium
JA[3]	Medium
JA[4]	Medium
JA[5]	Medium
JA[6]	Medium
JA[7]	Medium
JB[0]	Medium
JB[1]	Medium
JB[2]	Medium
JB[3]	Medium
JB[4]	Medium
JB[5]	Medium
JB[6]	Medium
SW_DATA_RATE	Medium
SW_DIN[0]	Medium
SW_DIN[1]	Medium
SW_DIN[2]	Medium
SW_DIN[3]	Medium
SW_DIN[4]	Medium
SW_DIN[5]	Medium
SW_DIN[6]	Medium
SW_DIN[7]	Medium

Figure 9. No input delay warnings — post-fix



no_output_delay

Name	Severity
Ports with no output delay but user has a false path constraint (18)	Medium
JC[0]	Medium
JC[1]	Medium
JC[2]	Medium
JC[3]	Medium
JC[4]	Medium
JC[5]	Medium
JC[6]	Medium
JC[7]	Medium
JD[0]	Medium
JD[1]	Medium
JD[2]	Medium
JD[3]	Medium
JD[4]	Medium
JD[5]	Medium
JD[6]	Medium
JD[7]	Medium
LED_BUSY	Medium
LED_DONE	Medium

Figure 10. No output delay warnings — post-fix

9. Final FPGA Results

9.1 Timing Summary

The implementation fully satisfies all user-defined timing constraints without any failing endpoints. Operating on the primary **32 MHz** clock domain, the setup timing achieves a Worst Negative Slack (WNS) of **+10.493 ns** across **11,863** total endpoints with a Total Negative Slack (TNS) of **0.000 ns**. Hold timing is closed with a Worst Hold Slack (WHS) of **+0.139 ns** (THS = **0.000 ns**), and pulse width constraints are met with a Worst Pulse Width Slack (WPWS) of **+2.633 ns** across **6,788** endpoints. These positive margins verify that no signal races or propagation setup violations exist in the fabric.

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 10.493 ns	Worst Hold Slack (WHS): 0.139 ns	Worst Pulse Width Slack (WPWS): 2.633 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 11863	Total Number of Endpoints: 11863	Total Number of Endpoints: 6788

All user specified timing constraints are met.

Figure 11. Design timing summary

9.2 Power Summary

Total estimated on-chip power consumption is **0.225 W**, maintaining a junction temperature of **26.0°C** with a thermal margin of **59.0°C (12.8 W)**. Dynamic power accounts for **0.128 W** (57% of total power), while device static leakage accounts for **0.098 W** (43%). The dynamic power profile is heavily dominated by the single instantiated PLL clock manager, which consumes **0.106 W** (83% of dynamic power). Core clock routing consumes **0.009 W** (7%), while active logic switching and internal signal routing consume **0.005 W** (4%) each. I/O driver switching power remains low at **0.003 W** (1%).

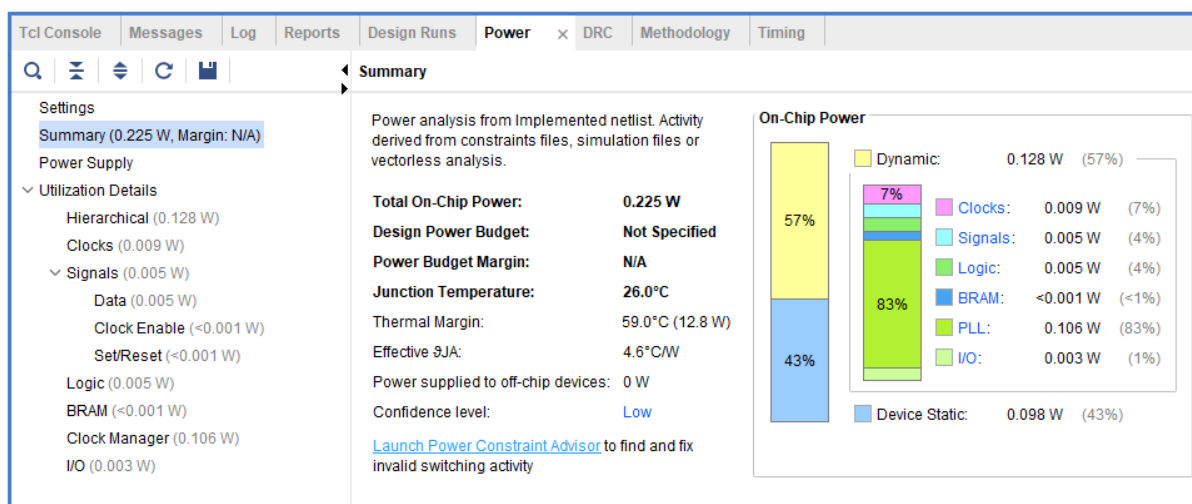


Figure 12. Power summary report

9.3 Utilization Summary

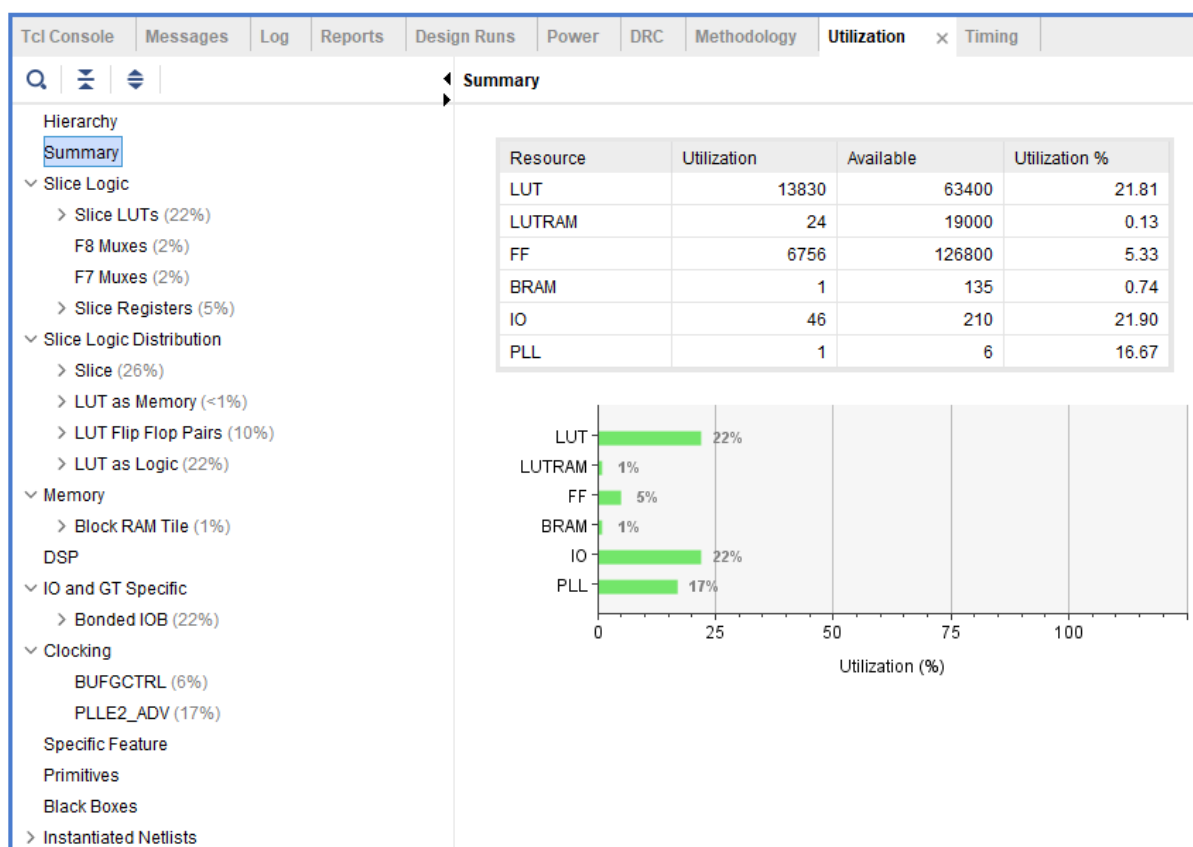


Figure 13. Utilization summary report

Resource	Used / Available	Utilization
Slice LUTs	13,830 / 63,400	21.81%
Slice Flip-Flops	6,756 / 126,800	5.33%
Bonded IOBs	46 / 210	21.90%
PLLs	1 / 6	16.67%
Block RAM Tiles	1 / 135	0.74%
Distributed LUTRAM	24 / 19,000	0.13%

The asymmetrical ratio of LUT usage (**21.81%**) compared to Flip-Flop usage (**5.33%**) is driven by the dynamic bit manipulation in `form_ppdu.v`. Converting the I/Q FIFOs to packed 2,048-bit vectors with indexed part-select (+:) generated large multiplexer structures and barrel-shifter steering networks in the combinational fabric. Additionally, using Distributed LUTRAM (24 bits) for asynchronous memory access in the capture

buffers absorbed combinational LUT primitives rather than dedicated BRAM blocks, keeping total Block RAM usage down to just **1 tile (0.74%)**.